MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, M229
PARTS FOR

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for one type of warhead designated as Warhead, 2.75 Inch Rocket, M229.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms & Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammo. Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ORDNANCE CORPS

921 8699	-	Warhead, 2.75 Inch Rocket, M229, Metal Parts Assembly
7548604	-	Pallet (Boxed Type) for Projectiles and Projectile Metal Parts

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. -Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Warhead Assemblies .-The warhead assemblies shall comply with all requirements specified on Drawing (dwg.) 9218699, and with all requirements specified in applicable specifications.

3.3 Metal Defects .-The components shall be free of surface defects, cracks, cracklike imperfections, pits, cold shuts, hot tears, blow holes, scaling and other similar defects.

3.4 Workmanship. -The components and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, unblended radii, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing phosphate coating or base material. All required markings shall be neat and sharply defined.

3.5 Salt Spray Test .-The finished assemblies, after painting, shall comply with an accelerated corrosion test as described in this document.

3.6 First Article Inspection. -This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product .-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization. A contractor shall not knowingly include defective units in a lot.

4.1.2 Inspections .-Unless otherwise specified in the contract, the following types of inspection(s) shall be conducted:

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a. First Article Inspection

b. Quality Conformance (production Lot Acceptance)Inspection

4.2 First Article Inspection

4.2.1 Submission.-All components and items submitted in the first article sample shall have been produced by the contractor or furnished by a sub-contractor and shall have been manufactured using the same production process, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and shall submit a record of this inspection with the sample, including statements of findings for tests and examinations, and certificates of conformance for materials (See 4.1.1c). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of' 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government. The first article sample shall consist of machined tensile specimens from two nose, two body, and two bases (as necessary to conduct the tests in 4.3.3.2, 4.3.3.4 and 4.3.3.5) along with five sets of the component parts (unpainted), five unpainted assemblies and five assemblies. See paragraph 6.4 of this specification for instructions for submission.

4.2.2 Inspections to be Performed.-First article assemblies, components and test specimens will be subjected by the Government to any or all of the examinations and tests specified in this detail specification and to any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure in the sample to comply with any of the stated requirements.

4.3 Quality Conformance (Production Lot Acceptance) Inspection

4.3.1 Lot Formation.-The term “inspection lot”, as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements, The sample selected shall represent only that quantity of units from which the sample was drawn and shall not

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be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions> and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot . All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with require-ments. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number.

4.3.2 Examination. -The examinations listed in this detail specification for Classification of Defects (and tests) shall be performed on inspection lots as defined in 4.3.1 of this speci-fication.

a. Sampling Plans .-Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that variables sampling plans in accordance with MIL-STD-414, or continuous sampling plans in accordance with MIL-STD-1235 may be used if requested by the contractor and approved by the Government.

b. Critical Defects .-Unless otherwise specified, in-spection for Critical Defects shall be 100 percent (%).

Major and Minor Defects. -When required, Major De-fects shall be inspected 100 percent. Except where 100 percent inspection is specified, inspection for Major and Minor Defects shall be to the individual AQL indicated in the Classification of Defects and Test paragraphs provided in this detail specifi-cation. The sampling plans for testing (See paragraph 4.3.3) shall be as specified.

d. Unlisted Defects.-See MIL-A-2550.

e. Disposition. Unless otherwise specified, the dispo-sition of defective units and rejected lots shall be in accordance with the requirements of MIL-STD-105.

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f. Instead of applying the AQL's designated herein to the individual defects, the group AQL's listed for each Major or Minor subparagraph in 4.3.2 may be utilized at the option of the contractor for those groups where a group AQL is assigned. Each group AQL applies to a Major or Minor group indicated in the subparagraphs of paragraph 4.3.2. When group AQL's are elected, they shall be utilized for all of the sub-paragraphs of paragraph 4.3.2.

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PARAGRAPH			SHEET 1 OF 1	DRAWING NUMBER 9218701
4.3.2.1	Nose Casting			COVERING DETAIL OF DWG. 9218699
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined			
MAJOR:				
101.	Hardness	4.3.3.1	3.1	4.4.1
102.	Tension	4.3.3.2	3.1	4.4.2
MINOR:	None Defined			

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PARAGRAPH				DRAWING NUMBER
4.3.2.2	Nose		SHEET 1 OF 1	9218701
				COVERING DETAIL OF DWG. 9218699
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined			
MAJOR:				
101.	Metal defective	0.40	3.3	Visual
MINOR:	None defined			

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PARAGRAPH			SHEET 1 OF 1	DRAWING NUMBER 9218700
4.3.2.3	Body Casting			COVERING DETAIL OF DWG. 9218699
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined			
MAJOR:				
101.	Hardness	4.3.3.3	3.1	4.4.3
102.	Tension	4.3.3.4	3.1	4.4.4
MINOR:	None Defined			

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PARAGRAPH				SHEET 1 OF 1	DRAWING NUMBER 9218700
4.3.2.4	Body				COVERING DETAIL OF DWG. 9218699
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA.	REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined				
MAJOR:					
101.	Metal Defective	0.40		3.3	Visual
MINOR:	None Defined				

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PARAGRAPH				DRAWING NUMBER
4.3.2.5	Base, before Machining		SHEET 1 OF 1	8889404
				COVERING DETAIL OF DWG. 9218699
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined			
MAJOR:				
101.	Tension Test	4.3.3.5	3.1	4.4.5
MINOR:	None Defined			

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PARAGRAPH				SHEET 1 OF 1	DRAWING NUMBER 8889404 COVERING DETAIL OF DWG. 9218699
4.3.2.6	Base				
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.		REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined				
MAJOR:	GROUP AQL 0.65%				
101.	Diameter inside at front	0.40		3.1	Plug, Taper
102.	Diameter inside	0.40		3.1	Plug, Taper
103.	Metal Defective	0.40		3.3	Visual
MINOR:	None Defined				

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PARAGRAPH		SHEET 1	DRAWING NUMBER
4.3.2.7	Warhead, 2.75" Rocket M229, Metal Parts Assy. before painting	OF 5	9218699 COVERING DETAIL OF DWG. N/A

CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL.				
1.	Hydrostatic Test	4.3.3.6	3.2	4.4.6
2.	Hydrostatic stamp missing	100%	3.2	Visual
3.	Air Test	4.3.3.7	3.2	4.4.7
4.	Air pressure stamp missing	100%	3.2	Visual
MAJOR:	GROUP AQL 2.5% (Except 124)			
101.	Concentricity of pitch diameter of fuze thread with outside diameter between the bourrelets including perpendicularity of front face	0.40	3.2	Concentricity thread plug with fixtured indicator
102.	Concentricity of counterbore at front of fuze thread with pitch diameter of thread	0.40	3.2	Concentricity thread plug

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PARAGRAPH		SHEET 2	DRAWING NUMBER
4.3.2.7	Warhead, 2.75" Rocket M229, Metal Parts Assy. before painting	OF 5	9218699 COVERING DETAIL OF DWG. N/A

CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
103.	Concentricity of diameter at rear of fuze thread undercut (if applicable) with pitch diameter of threads	0.40	3.2	Concentricity thread plug
104.	Concentricity of ogive with outside diameter between the bourrelets.	0.40	3.2	Fixtured indicator
105.	Concentricity of base at rear of thread w/pitch diameter of base thread	0.40	3.2	Plug, concentricity thread w/fixtured indicator
106.	Concentricity of base at rear of threads with outside diameter between the bourrelets	0.40	3.2	Fixtured indicator
107.	Concentricity of the rear bourrelet w/outside diameter between the bourrelets	0.40	3.2	Fixtured indicator

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PARAGRAPH 4.3.2.7 — Warhead, 2.75" Rocket M229, Metal Parts Assy. before painting — SHEET 3 OF 5 — DRAWING NUMBER 9218699 — COVERING DETAIL OF DWG. N/A

CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
108.	Concentricity of the forward bourrelet w/outside diameter between the bourrelets	0.40	3.2	Fixtured indicator
109.	Motor body seat not perpendicular to outside diameter between the bourrelets	0.40	3.2	Fixtured indicator
110.	Length of fuze thread (if applicable)	0.40	3.2	Thread Plug
111.	Diameter of counterbore at front fuze thread	0.40	3.2	Set ring w/indicator
112.	Diameter of thread undercut (if applicable)	0.40	3.2	Set ring w/indicator
113.	Diameter of rear of fuze thread (rear of thread undercut if applicable)	0.40	3.2	Set ring w/indicator
114.	Diameter of nose between the forward bourrelet and ogive	0.40	3.2	Adjustable snap
115.	Diameter of base at rear of thread	0.40	3.2	Adjustable snap

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PARAGRAPH 4.3.2.7 — Warhead, 2.75" Rocket M229, Metal Parts Assy. before painting — SHEET 4 OF 5 — DRAWING NUMBER 9218699 — COVERING DETAIL OF DWG. N/A

CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR REQMT. PARA. REF.	REQMT. PARA.	METHOD OF INSP.
116.	Thickness of wall between the forward bourrelet & ogive	0.40	3.2	Fixtured indicator
117.	Thickness of wall between the bourrelets	0.40	3.2	Fixtured indicator
118.	Variation of wall thickness excessive	0.40	3.2	Fixtured indicator
119.	Profile of ogive	0.40	3.2	Special template w/flush pins
120.	Thickness through base	0.40	3.2	Fixture gage
121.	Metal defective	0.40	3.3	Visual
122.	Length from front face to motor seat	0.40	3.2	Fixture gage
123.	Length from motor seat to rear of base	0.40	3.2	Flush pin gage
124.	Inside diameter gap inspection	4.3.3.8	3.2	Visual w/borescope
MINOR:	GROUP AQL 2.5%			
201.	Depth of counterbore at front	0.65	3.2	Flush pin gage
202.	Depth of bottom of dia. at rear of thread undercut (if applicable)	0.65	3.2	Length gage

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PARAGRAPH		SHEET	DRAWING NUMBER
4.3.2.7	Warhead, 2.75" Rocket M229, Metal Parts Assy. before painting	5 OF 5	9218699
			COVERING DETAIL OF DWG. N/A

CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
203.	Improper radii or chamfer	0.65	3.2	Visual
204.	Improper surface finish	0.65	3.2	Visual
205.	Location of rear thread	0.65	3.2	Flush pin gage
206.	Gap on OD between nose and body excessive	0.65	3.2	Feeler gage
207.	Evidence of poor workmanship other than that covered under Major category above	0.65	3.4	Visual

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PARAGRAPH		SHEET 1	DRAWING NUMBER 9218699
4.3.2.8	Warhead, 2.75" Rocket, M229, Metal Parts Assy.	OF 2	COVERING DETAIL OF DWG. N/A

CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
CRITICAL:	None Defined			
MAJOR:	GROUP AQL 1.00% (Except 109 and 110)			
101.	Pitch dia. of fuze thread	0.40	3.2	Thread plug
102.	Minor dia. of fuze thread, max.	0.40	3.2	Plain cylindrical plug
103.	Pitch dia. of rear thread	0.40	3.2	Thread ring
104.	Major dia. of rear thread, min.	0.40	3.2	Adjustable snap
105.	Thread damaged or not full	0.40	3.2	Visual
106.	Foreign matter in cavity	0.40	3.2	Visual
107.	Dia. of rear bourrelet, max.	0.40	3.2	Ring
108.	Dia. of forward bourrelet, max.	0.40	3.2	Ring
109.	Salt Spray	4.3.3.9	3.5	4.4.8
110.	Drop Test	4.3.3.10	3.2	4.4.9

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PARAGRAPH			SHEET 2	DRAWING NUMBER 9218699
4.3.2.8	Warhead, 2.75" Rocket, M229, Metal Parts Assy.		OF 2	COVERING DETAIL OF DWG. N/A
CATEGORY	INSPECTION CHARACTERISTICS	AMT. OF INSP. AQL, 100% OR PARA. REF.	REQMT. PARA.	METHOD OF INSP.
MINOR:	GROUP AQL 2.50%			
201.	Protective coating inadequate or damaged	0.65	3.2	Visual
202.	Paint in cavity inadequate	0.65	3.2	Visual
203.	Marking missing, incorrect, or illegible	0.65	3.2	Visual

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4.3.3 Testing.

4.3.3.1 Hardness Test on Nose - Major Defect.-To insure compliance with the mechanical properties requirements, each nose casting from a sample shall be subjected to a hardness test with a Brinell Hardness Tester. The sample, 10 percent minimum representative of extreme furnace environment, shall be selected from each heat treat batch when a batch type furnace is used for final heat treatment. When a batch type furnace is used for final heat treatment, the following shall be substituted for a batch. For identification purposes, any quantity of nose castings, but not to exceed 4 hours production of the final heat treatment that have been subjected to final heat treatment in the same furnace, at the same temperature for the same length of time or conveyor speed, shall be considered a heat treatment batch. The heat treatment batch thus constituted shall form the basis for selecting sample castings for the tension test hereinafter prescribed. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

4.3.3.2 Tension Test on Nose - Major Defect.-Two (2) nose castings from each heat treatment batch, one of the hardest and one of the softest, as determined by 4.3.3.1, shall be subjected to this test. Failure of one or more tension test specimens to comply with the physical requirements of the applicable drawing shall be cause for rejection of the batch represented by the specimen. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.3 Hardness Test on Body - Major Defect.-To insure compliance with the mechanical properties requirements, each nose casting from a sample shall be subjected to a hardness test with a Brinell Hardness Tester. The sample, 10 percent minimum, representative of extreme furnace environment, shall be selected from each heat treat batch when a batch type furnace is used for final heat treatment. When a continuous type furnace is used for final heat treatment, the following shall be substituted for a batch. For identification purposes, any quantity of body castings, but not to exceed 4 hours' production of the final heat treatment furnace, that have been subjected to final heat treatment in the same furnace, at the same temperature, for the same length of time or conveyor speed, shall be considered a heat treatment batch. The heat treatment batch thus constituted shall form the basis for selecting sample castings for the tension test hereinafter prescribed. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

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4.3.3.4 Tension Test on Body - Major Defect.-Two (2) body castings from each heat treatment batch, one of the hardest and one of the softest, as determined by 4.3.3.3, shall be subjected to this test. Failure of one or more tension test specimens to comply with the physical requirements of the applicable drawing shall be cause for rejection of the batch represented by the specimen. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.5 Tension Test on Base - Major Defect.-Two (2) bases from each heat treatment or stress-relieved batch of 1,000 or fraction thereof, shall be subjected to the tension test in accordance with 4.4.5, to determine compliance with the mechanical properties requirements of the applicable drawing. When steel bases are used, two (2) bases from each heat treatment or stress-relieved batch of 5,000 or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply shall be cause for rejection of the batch represented by the specimen. The test shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4.

4.4.3.5.1 Retest .-When one or more tension test specimens fail to comply with the yield strength or elongation requirements of the applicable component drawing, the rejected batch may, at the discretion of the contractor, be re-heat-treated, or new samples of a higher or lower hardness value or both may be selected (if this was the basis for selecting samples). A new sample may also be cut from the same test casting if a visible defect was present in the test bar. When maximum and minimum hardness values have been determined by compliance with the tensile test specimens with the mechanical properties requirements of the component drawing, these values shall be used as a basis for accepting or rejecting each nose casting within that batch. All castings within the batch whose hardness values fall outside the established range shall be rejected subject to reheat treatment and resubmission for test.

4.3.3.6 Hydrostatic Test - Critical Defect.-Each metal parts assembly shall be subjected to the required internal hydrostatic pressure to determine compliance with requirements on dwg. 9218699. Failure of any assembly to comply shall be cause for rejection of the assembly. Metal parts assemblies which oomply shall be stamped with the letter “H” as indicated on the assembly drawing. The test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3.4

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4.3.3.7 Air Pressure - Critical Defect.-Each body assembly that has passed the hydrostatic pressure test, as indicated by the letter "H" stamped on the body, shall, after completion of the drying cycle, be subjected to submerged air pressure test. Any body assembly failing to comply with the requirements specified on dwg. 9218699, as evidenced by air bubbles rising to the surface of the water, shall be rejected. Air bubbles caused by entrapped air on the outside of the shell shall not be cause for rejection of the body assembly. Body assemblies which comply with the requirements of dwg. 9218699 shall be stamped with the letter "A" as indicated on the assembly drawing. The test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4.

4.3.3.8 ID Gap Inspection - Major Defect.-A sample of fifty (50) unpainted assemblies, randomly selected from each lot, shall be inspected for evidence of gap (any discernible separation between nose and body section on ID at brazed joint). If one or more of the assemblies inspected show evidence of one continuous gap over an arc in excess of five (5) degrees, or the total of all the gaps discernible equal an arc in excess of' ten (10) degrees, the lot represented by the sample shall be rejected.

4.3.3.9 Salt Spray Test - Major Defect.-The sample size for this test shall be as follows: During production, one (1) warhead from each day's production shall be subjected to the salt spray test as specified in 4.4,8, until five (5) consecutive days' production have successfully passed the test . Sampling shall then be reduced to one (1) sample twice per week. If, upon reduced sampling, one (1) failure occurs, corrective action shall be taken and sampling shall revert to one specimen from each day's production until five (5) consecutive days' production have again successfully passed the test. If two (2) consecutive failures occur after the contractor has been notified to take corrective action in either the original sampling plan or the reduced sampling plan, inspection for acceptance will be stopped until corrective action has been taken. Any specimen (warhead) which does not comply with any of the following test criteria shall be considered a failure. The specimen, when subjected to the salt spray test specified in 4.4.8, shall show no more than 1/8 inch creepage, blistering or loss of adhesion of the paint as measured from the scribe mark. At all other points there shall be no more than a trace of film failure, such as blistering or corrosion, and not more than twenty (20) scattered blisters, none larger than 3/16 inch in diameter on the painted surface of the warhead. This is intended that the maximum amount of corrosion in the salt spray test be comparable to photograph 8-1 in ASTM Standard

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Method of Evaluation Degree of Resistance to Rusting Obtained with Paint or Iron and Steel Surfaces, D610-43. This test shall be performed as specified in 4.4.8, using equipment in accordance with 4.3.4.

4.3.3.10 Drop Test - Major Defect.-Beginning With the first lot and continuing until five (5) consecutive lots have been tested, a sample from each lot shall be subjected to a drop test as specified in 4.4.9, using equipment in accordance with 4.3.4. Upon satisfactory results from five (5) consecutive lots, the contractor may reduce the testing to a sample from one randomly selected lot per month. The contractor shall return to the five-lot requirement upon failure of any lot. The sample shall consist of 23 randomly selected painted assemblies which have passed all other requirements of this specification. The lot represented by the sample shall be rejected if two (2) or more warheads fail to comply with the inspection criteria. If only one (1) warhead fails, a retest is permissible as follows:

For retest, a new sample consisting of 46 warheads randomly selected from the lot shall be subjected to the drop test in accordance with 4.4.9. If one or more warheads fail to comply with the inspection criteria, the lot shall be rejected.

Inspection Criteria. - The warheads shall be subjected to, and comply with, a hydrostatic and air test in accordance with 4.3.3.6 and 4.3.3.7. There shall be no visible evidence of cracklike discontinuities on the warheads after the Hydrostatic and Air Tests. This is considered to be a destructive test.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.4 (See 6.3).

4.4 Test Methods and Procedures

4.4.1 Hardness Test on Nose - (See 4.3.3.1).-After final heat treatment, the hardness test shall be performed on the nose forward of the bourrelet. Procedures for this test are as prescribed by Federal Test Method Standard No. 151, Method 242.1.

4.4.2 Tension Test on Nose - (See 4.3.3.2).-One (1) tension test specimen conforming to a standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each sample. The criteria for determining yield strength and elongation shall

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be as prescribed by Federal Test Method Standard No. 151, Method 211.1.

4.4.3 Hardness Test on Body - (See 4.3.3.3).-After final heat treatment, the hardness test shall be performed on the body forward of the bourrelet. Procedures for this test are as prescribed by Federal Test Method Standard No. 151, Method 242.1.

4.4.4 Tension Test on Body - (See 4.3.3.4).-One tension test specimen conforming to a standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each sample. The criteria for determining yield strength and elongation shall be as prescribed by Federal Test Method Standard No. 151, Method 21101.

4.4.5 Tension Test on Base (See 4.3.3.5) .-From the samples provided in accordance with 4.3.3.5, one (1) tensile test specimen conforming to a standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from the bottom of each base. The criteria for determining yield strength and elongation shall be as prescribed by Federal Test Method Standard No. 151$_3$ Method 211.1$_0$

4,4.6 Hydrostatic Test (See 4.3.3.6).-The metal parts assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached, after which the pressure will be held for a minimum of 5 seconds. The equipment shall be calibrated prior to the start of each day's operation, at intervals of not over 2 hours of continuous operation, and prior to restarting the operation after a layoff of 1 hour of more.

4.4.7 Air Pressure Test - (See 4.3.3.7).-Body assemblies that have passed the hydrostatic test shall be mounted in an approved fixture in such a manner that the body assembly will be completely submerged in water. The air pressure shall be applied and held for a minimum of 15 seconds.

4.4.8 Salt Spray Test (See 4.3.3.9).-An accelerated corrosion test shall be performed utilizing the equipment and procedures prescribed by the Federal Test Method Standard No. 151, Method 811, A 20 percent concentration of sodium chloride solution shall be used, and the length of the test shall be for 144 hours. Actual production warhead assemblies only shall be subjected to this test. Test panels

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shall not be substituted for the warhead assemblies. The test specimen shall be air dried for a minimum of 96 hours before testing. The painted specimen shall be scored through using a sharp knife. The specimens shall then be exposed to the 20 percent salt spray as specified in Federal Test Method Standard No. 151, Method 811.

4.4.9 Drop Test.-Warheads from a sample as provided in accordance with 4.3.3.10 shall be dropped from a height of 5 feet onto a steel plate having a minimum thickness of 3 inches and a brinell hardness of at least 207. The steel plate shall be solidly supported in a horizontal plane over its entire bearing area by a minimum thickness of 24 inches of concrete. The steel plate shall have a smooth, flat surface and shall have a length and width of at least 36 inches. The plate shall be surrounded on all four sides by a heavy steel enclosure of sufficient height and strength to contain the warhead during rebounds. The 5 foot height necessary to perform this test can be obtained by using a special steel tower or derrick. The warheads will be dropped individually with their longitudinal axis at 45 degrees to the horizontal, base down. The warhead shall be inert loaded to conform to the weight and center of gravity of a loaded warhead. An inert loaded 423 fuze shall be attached to the inert warhead. The combined weight of the inert warhead and fuze shall be 16.75 ± 0.2 lbs. The center of gravity if 14.25 ± 0.5 inch from the forward end of the fuze.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking - Level C.-Warheads shall be packed and marked in accordance with drawing 7548604.

6. NOTES

6.1 Intended Use .-See paragraph 1.1 of this specification.

6.2 Ordering Data.- Procurement documents shall specify the following:

a. Title, number and date of this and item detail specifications.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167, except when propellant or explosive lots are involved.

c. Quantity required and delivery schedules.

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d. Serialization requirements, if applicable.

e. Production lot acceptance, if other than specified in Section 4 of this specification.

f. First Article Sample requirements, if other than specified in Section 4 of this specification.

g. Packaging requirements, if other than specified in Section 5 of this specification.

h. Inspection equipment design requirements (See 4.3.4 and 6.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) Designs and Contractor Designs. SIE Designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the Contractor. The Contractor need not furnish any design when a complete Government SIE Design is provided. Unless otherwise specified, however, the Contractor may submit alternate or modified Contractor Designs of SIE in accordance with paragraphs 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE Designs .-SIE Designs may consist of any or the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs .-Contractor Designs are required for all inspection equipment for which SIE Designs are not specified, and may include commercial equipment which the Contractor proposed to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor Designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings

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or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any specific fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423), The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification, paragraph, and defect numbers from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-R, Dover, N.J. 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Sample.-Instructions as to the submission and location for testing and evaluation of the first article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal, and instructions will be issued accordingly. All inquiries should be forwarded to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-I-Q
Dover, New Jersey 07801

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6.5 Definitions.

6.5.1 Contractor.-Unless otherwise specified, the word "Contractor", used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

6.6 Supersedes Picatinny Arsenal Purchase Description, PA-PD2913 dated 4 January 1967 and its associated Engineering Orders and Notice of Revision.

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Preparing Activity:
ARMY - PA

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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